

Interview Questions for Android Developer

1. What is the difference between `Activity` and `Fragment` in Android?

ANSWER: An `Activity` represents a single screen with a user interface, while a `Fragment` represents a reusable portion of the UI that can be embedded in an activity.

2. How do you communicate between two fragments?

ANSWER: You can communicate between two fragments using a shared ViewModel or by passing data through the parent activity using interfaces.

3. What is an Intent in Android?

ANSWER: An Intent is a messaging object used to request an action from another app component, such as starting an activity or a service.

4. Explain the lifecycle of an `Activity`.

ANSWER: Key lifecycle methods include `onCreate()`, `onStart()`, `onResume()`, `onPause()`, `onStop()`, `onDestroy()`, and `onRestart()`.

5. What is the difference between `onCreate()` and `onStart()`?

ANSWER: `onCreate()` is called when the activity is first created, and it initializes the activity. `onStart()` is called just before the activity becomes visible to the user.

6. How do you handle configuration changes in Android?

ANSWER: You can handle configuration changes by saving the state in `onSaveInstanceState()` and restoring it in `onRestoreInstanceState()` or by using a retained fragment or ViewModel.

7. What is an `AsyncTask` and when would you use it?

ANSWER: `AsyncTask` is used to perform background operations and publish results on the UI thread without having to manipulate threads and handlers.

8. Explain the concept of `ViewModel` in Android.

ANSWER: `ViewModel` is a class designed to store and manage UI-related data in a lifecycle-conscious way, surviving configuration changes like screen rotations.

9. What is the difference between `Service` and `IntentService`?

ANSWER: A `Service` runs on the main thread and can perform long-running operations, while an `IntentService` runs on a separate worker thread and stops itself when the work is done.

10. How do you perform data binding in Android?

ANSWER: Data binding in Android can be done using the Data Binding Library, which allows you to bind UI components to data sources in your layout XML files.

11. What is an `Adapter` in Android?

ANSWER: An `Adapter` acts as a bridge between an `AdapterView` (like `ListView`, `GridView`, `RecyclerView`) and the underlying data for that view.

12. How do you use a `RecyclerView`?

ANSWER: To use a `RecyclerView`, you need to define a layout manager, an adapter, and a ViewHolder to efficiently display and manage a large dataset.

13. Explain the difference between `Serializable` and `Parcelable`.

ANSWER: `Parcelable` is a faster and more efficient serialization mechanism for passing data between Android components compared to `Serializable`.

14. How do you manage the Android activity back stack?

ANSWER: You manage the back stack using intents with appropriate flags, the `finish()` method, and the `FragmentManager` for fragments.

15. What is a `ContentProvider`?

ANSWER: A `ContentProvider` manages access to a structured set of data and provides mechanisms for data sharing between applications.

16. How do you handle permissions in Android?

ANSWER: Permissions are handled by declaring them in the AndroidManifest.xml and requesting them at runtime using the `requestPermissions()` method.

17. What are the different types of storage available in Android?

ANSWER: Android provides several storage options, including SharedPreferences, Internal Storage, External Storage, and SQLite databases.

18. What is the purpose of the `manifest` file in an Android project?

ANSWER: The `AndroidManifest.xml` file provides essential information about your app to the Android system, including permissions, components, and application metadata.

19. Explain the concept of `BroadcastReceiver`.

ANSWER: `BroadcastReceiver` is a component that responds to system-wide broadcast announcements or intents.

20. How do you ensure your app is responsive and avoids ANR (Application Not Responding) errors?

ANSWER: To avoid ANR errors, perform long-running operations on background threads using `AsyncTask`, `Thread`, `HandlerThread`, or other concurrency mechanisms.

21. What is the purpose of `Handler` and `Looper` in Android?

ANSWER: `Handler` and `Looper` are used to manage threads, allowing you to send and process messages and runnables on a particular thread's message queue.

22. How do you update the UI from a background thread?

ANSWER: You update the UI from a background thread by using `Handler`, `runOnUiThread()`, or other methods like `LiveData` with `ViewModel`.

23. What is the `JobScheduler` API?

ANSWER: `JobScheduler` is an API that allows you to schedule jobs to be executed in the background based on certain conditions, like network availability or charging status.

24. Explain the usage of `Room` database.

ANSWER: `Room` is a persistence library that provides an abstraction layer over SQLite to allow fluent database access while harnessing the full power of SQLite.

25. What is `Dagger` in Android development?

ANSWER: `Dagger` is a dependency injection library that helps manage dependencies in an Android application.

26. How do you handle multiple screen sizes and resolutions in Android?

ANSWER: You handle multiple screen sizes and resolutions by using flexible layouts, drawable resources for different densities, and dimension resources.

27. What are the benefits of using `ConstraintLayout`?

ANSWER: `ConstraintLayout` allows for complex layouts with a flat view hierarchy, improving performance and flexibility.

28. What is a `ViewHolder` pattern in Android?

ANSWER: The `ViewHolder` pattern is used to improve the performance of `ListView` and `RecyclerView` by avoiding unnecessary `findViewById()` calls.

29. Explain the purpose of `ProGuard`.

ANSWER: `ProGuard` is a tool that helps to shrink, obfuscate, and optimize your code, reducing the size of your APK and protecting it from reverse engineering.

30. How do you use `LiveData` and `ViewModel` in Android?

ANSWER: `LiveData` is used to observe data changes, and `ViewModel` is used to manage UI-related data in a lifecycle-conscious way.

31. What is a `Coroutine` in Kotlin?

ANSWER: A `Coroutine` is a concurrency design pattern used in Kotlin to simplify code that executes asynchronously.

32. How do you use `Retrofit` for network operations in Android?

ANSWER: `Retrofit` is a type-safe HTTP client for Android and Java, used to handle REST API calls by creating a service interface and defining HTTP methods.

33. What is `MVP` and `MVVM` architecture in Android?

ANSWER: `MVP` (Model-View-Presenter) and `MVVM` (Model-View-ViewModel) are architectural patterns that separate concerns to make the code more modular, testable, and maintainable.

34. How do you implement a `SplashScreen` in Android?

ANSWER: A `SplashScreen` can be implemented using a separate activity with a timer, or using `WindowBackground` and theming.

35. What is `Navigation Component` in Android?

ANSWER: `Navigation Component` is a part of Jetpack that helps you implement navigation, including handling fragment transactions, deep linking, and back stack management.

36. How do you manage dependencies in an Android project?

ANSWER: Dependencies are managed using Gradle, where you define dependencies in the `build.gradle` file.

37. What is `Lint` in Android development?

ANSWER: `Lint` is a static analysis tool used to check your Android project for potential bugs, code inefficiencies, and other issues.

38. How do you use `WorkManager` in Android?

ANSWER: `WorkManager` is used to schedule and manage deferrable, guaranteed background work, such as periodic tasks or tasks that need to be run under specific conditions.

39. What is `DataBinding` in Android?

ANSWER: `DataBinding` is a library that allows you to bind UI components in your layouts to data sources in your app using a declarative format.

40. How do you optimize battery usage in an Android app?

ANSWER: Optimizing battery usage involves minimizing background operations, using `JobScheduler` or `WorkManager` for deferred tasks, and using battery-efficient APIs.

41.What is a `Snackbar` in Android?

ANSWER: A `Snackbar` provides a lightweight feedback about an operation, showing a brief message at the bottom of the screen.

42. How do you handle runtime permissions in Android?

ANSWER: Handle runtime permissions by checking for permissions at runtime using `checkSelfPermission()`, requesting permissions using `requestPermissions()`, and handling the result in `onRequestPermissionsResult()`

43.What is a `ViewModelProvider` in Android?

ANSWER: `ViewModelProvider` is a utility class that provides ViewModels to your activity or fragment, ensuring they are scoped to the lifecycle of the component.

44.Explain the use of `SharedPreferences`.

ANSWER: `SharedPreferences` is used to store key-value pairs of primitive data types, which can be used for storing user preferences and settings.

45. How do you secure sensitive data in an Android app?

ANSWER: Secure sensitive data using encryption, the Android Keystore system, and avoiding hardcoding sensitive information in the app.

46.What is `Jetpack Compose`?

ANSWER: `Jetpack Compose` is a modern toolkit for building native Android UI, allowing you to create UIs using a declarative approach.

47. How do you debug an Android application?

ANSWER: Debug an Android application using tools like Android Studio's debugger, Logcat, breakpoints, and profiling tools.

48. What are `Broadcast Intents`?

ANSWER: `Broadcast Intents` are used to broadcast events system-wide or within the app, which can be received by `BroadcastReceivers`.

49. How do you implement localization in an Android app?

ANSWER: Implement localization by providing string resources in different languages in the `res/values` folder, and using resource qualifiers.

50. What is the `Paging` library in Android?

ANSWER: The `Paging` library helps load and display large data sets by dividing the data into pages, loading data gradually and efficiently.